Unit II :: Mathematical Typesetting and Technical Elements in LaTeX

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Syllabus

Unit II: Mathematical Typesetting and Technical Elements in LaTeX

Math mode: inline and display modes; math fonts: mathcal, mathbb, mathds, mathfrak; superscripts and subscripts, pre-defined functions, fractions, roots, Greek letters; integrals, summations and products; mathematical grids: multi-line formulae, gathering system of equations, aligning system of equations, matrices; LaTeX packages for Physics: siunitx, physics; floats in LaTeX: table and figure environments; typesetting tables using tabular; formatting figures using includegraphics.

1 Overview of math mode in LaTeX

In LaTeX, the typesetting of mathematical expressions requires entering a dedicated operating environment known as *math mode*. In math mode, LaTeX does the following

- Treats alphabet symbols as mathematical variables.
- Typesets letters in italics by default.
- Ignores standard whitespace characters.
- Adjusts spacing between symbols dynamically based on their semantic classification, such as operators, relations and delimiters.

1.1 Inline math mode

Inline math mode is used to embed short mathematical expressions and variables directly within a sentence without disrupting the paragraph flow. We can typeset a mathematical expression in inline mode using any of the following syntax:

1. Enclosing the expression within `$...$`, for example `$F=ma$`.
2. Enclosing the expression within `\(...\)`, for example `\(p=mv\)`.
3. Placing the expression within the `math` environment, for example
```
\begin{math}
    U=mgh
\end{math}
```

For instance, the following code contains all of the above three syntax

```
In classical mechanics, an object's translational motion and energetic state
near Earth's surface can be characterized by Newton's second law of motion $F=
ma$, linear momentum \(p=mv\), and gravitational potential energy
\begin{math}
  U=mgh
\end{math}.
```

which produces the following output:

In classical mechanics, an object's translational motion and energetic state near Earth's surface can be characterized by Newton's second law of motion $F = ma$, linear momentum $p = mv$, and gravitational potential energy $U = mgh$.

1.2 Display math mode

Display math mode is used for prominent, complex or multi-line equations that require dedicated vertical separation, centered alignment and full symbol scaling.

For display mode, we have two types of syntax:

1. *Un-numbered display syntax*:
   (a) Enclosing the expression within `$$...$$`, for example `$$F=ma$$`.
   (b) Enclosing the expression within `\[...\]`, for example `\[p=mv\]`.
   (c) Placing the expression within the `displaymath` environment, for example
   ```
   \begin{displaymath}
       U=mgh
   \end{displaymath}
   ```
2. *Numbered display syntax*: Placing the expression within the `equation` environment, in which case the equation will also be given a number. For example
   ```
   \begin{equation}
       T=\frac{1}{2}mv^2
   \end{equation}
   ```

For instance, the following code contains all of the above three syntax

```
In classical mechanics, an object's translational motion and energetic state
near Earth's surface can be characterized by Newton's second law of motion $$F=
ma$$ linear momentum \[p=mv\] gravitational potential energy
\begin{displaymath}
  U=mgh
\end{displaymath}
and kinetic energy
\begin{equation}
  T=\frac{1}{2}mv^2
\end{equation}
```

which produces the following output:

In classical mechanics, an object's translational motion and energetic state near Earth's surface can be characterized by Newton's second law of motion

$$F = ma$$

linear momentum

$$p = mv$$

gravitational potential energy

$$U = mgh$$

and kinetic energy

$$T = \frac{1}{2}mv^2 \tag{1}$$

2 Mathematical typography in LaTeX

2.1 Superscripts and subscripts

Superscripts and subscripts are commonly used to represent exponents (such as 3^2, x^{-3}) and indexed quantities (such as m_1, K_0) respectively. In LaTeX, superscripts are typeset using the *caret* symbol (`^`) while subscripts are typeset using the *underscore* symbol (`_`).

If a superscript or subscript contains more than a single character, the entire exponent or index must be enclosed in braces `{ }`.

For instance, the following code contains expressions with exponents and indices

```
$$ax^2+bx+c=0$$

$$f(x)=e^{-ax^2}$$

$$m_1=5, m_{12}=12$$
```

which produces the output

$$a x^2 + b x + c = 0$$

$$f(x) = e^{-a x^2}$$

$$m_1 = 5, m_{12} = 12$$

2.2 Fractions and roots

2.3 Pre-defined functions and operators

2.4 Summation, product and integral operators

2.5 Greek letters

2.6 Mathematical fonts